

感知机

给定输入 x , 权重 w 和偏移 b , 输出:

$$o = \sigma(w^T x + b), \quad \sigma(x) = \begin{cases} 1, & \text{if } x > 0 \\ 0, & \text{otherwise} \end{cases}$$

二分类: 1 或 -1.

回归: 输出实数.

Softmax: 回归输出概率

训练: 初始化 $w=0, b=0$.

循环:

if $y_i(w^T x_i + b) \leq 0$ 则 (分类错误),

$$w := w + y_i x_i, \quad b := b + y_i.$$

end if

直到全部正确.

等价于使用批量大小为1的梯度下降, 损失函数 $J(y, x, w) = \max(0, 1 - y w^T x)$
分类对硬新, 错误则更新

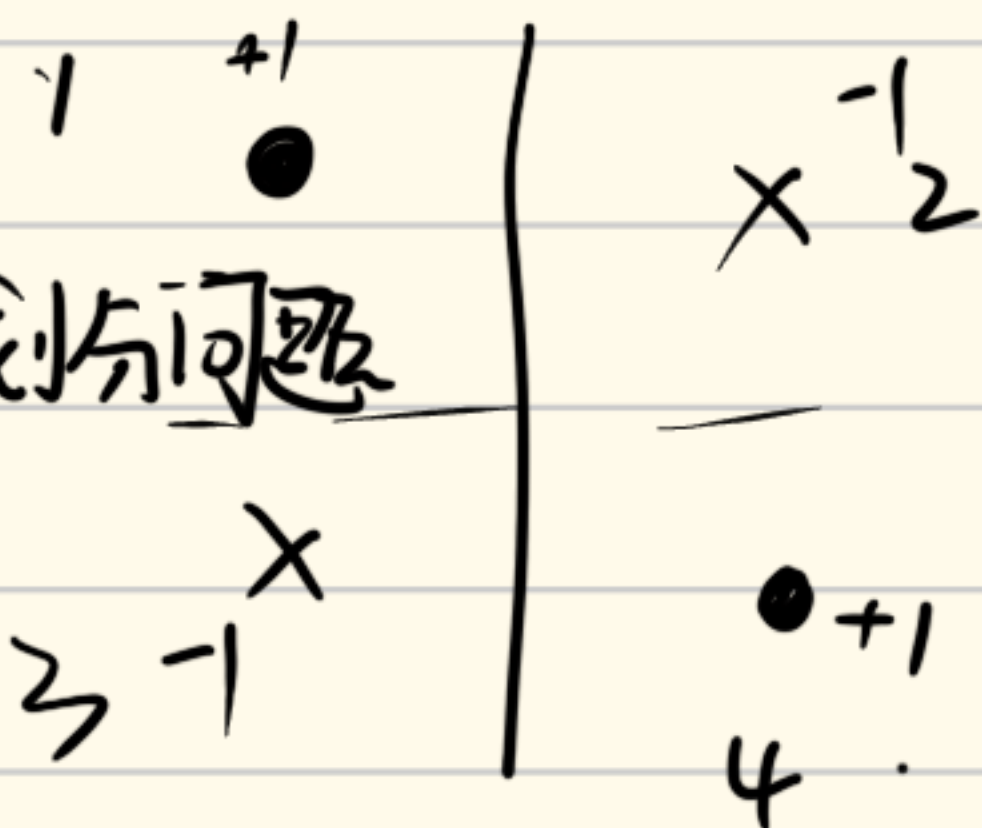
收敛定理: 数据在半径 r 内, 余量 P 分为两类

$$y(w^T x + b) \geq P$$

对于 $\|w\|^2 + b^2 \leq 1$, 保证 $\frac{r^2 + 1}{P^2}$ 步后收敛

问题: 异或问题:

感知机只能解决线性划分问题



多层感知机

	1	2	3	4
——	+	-	+	-
- - -	+	+	-	-
和.	+	-	-	+

输出层: $o_1 \quad o_2 \quad o_3 \quad o_4$

隐藏层: $h_1 \quad h_2 \quad h_3 \quad h_4 \quad h_5$
(大小为超层)

输入层: $x_1 \quad x_2 \quad x_3 \quad x_4$

单隐藏层—单分类:

输入 $x \in \mathbb{R}^n$, 隐藏 $W_1 \in \mathbb{R}^{m \times n}$. 输出 $u_2 \in \mathbb{R}^m$

$$h = \sigma(W_1 x)$$

$$o = w_2^T h + b_2$$

若无 σ , 则 $h = W_1 x$

$o = w_2^T h = w_2^T W_1 x$, 仍为线性. 所以需要激活函数

最经典的: $\text{sigmoid}(x) = \frac{1}{1+e^{-x}}$ (很软)

$$\tanh(x) = \frac{1-e^{-2x}}{1+e^{-2x}}$$

$\text{ReLU}(x) = \max(x, 0)$ (rectified linear unit) 计算廉价

多类分类: $y_1, y_2, \dots, y_k = \text{softmax}(o_1, \dots, o_k)$

$$h = \sigma(W_1 x)$$

$$o = W_2^T h$$

$$y = \text{softmax}(o)$$

每层都有4/降维

多隐藏层: $h_i = \sigma(W_i h_{i-1})$ 超参: 隐藏层数.

$$o = W_n h_{n-1}$$

每个隐藏层大小.

$$y = \text{softmax}(o)$$

得到 W 的方法: 反向传播

前向模式:

$$z_1 = W_1 x$$

$$h_1 = \sigma(z_1)$$

$$z_2 = W_2 h_1$$

$$h_2 = \sigma(z_2)$$

$\vdots$

$$z_n = W_n h_{n-1}$$

$$h_n = \sigma(z_n)$$

则梯度为 $\frac{\partial z}{\partial h_n} \cdot \frac{\partial h_n}{\partial z_n} \cdot \frac{\partial z_n}{\partial h_{n-1}} \cdot \frac{\partial h_{n-1}}{\partial z_{n-2}} \dots \frac{\partial h_2}{\partial z_2} \cdot \frac{\partial z_2}{\partial h_1} \cdot \frac{\partial h_1}{\partial z_1} \cdot \frac{\partial z_1}{\partial x}$

对于第 i 层, 梯度下降为 $W_i \leftarrow W_i - \eta \frac{\partial J}{\partial W_i}$

$$\frac{\partial J}{\partial W_i} = \frac{\partial J}{\partial z_i} \frac{\partial z_i}{\partial W_i} \quad \because z_i = W_i h_{i-1} \quad \text{则 } \frac{\partial z_i}{\partial W_i} = h_{i-1}^T$$

例: 2层MLP

$$\begin{cases} z_1 = W_1 x \\ h_1 = \sigma(z_1) \\ z_2 = W_2 h_1 \\ \hat{y} = z_2 \end{cases} \quad J = \frac{1}{2} \| \hat{y} - y \|^2 \quad \frac{\partial J}{\partial z_1} h_{i-1}^T$$

$$x = \begin{bmatrix} 0.4 \\ 0.7 \end{bmatrix} \quad W_1 = \begin{bmatrix} 0.2 & 0.5 \\ 0.1 & 0.3 \end{bmatrix} \quad W_2 = \begin{bmatrix} 0.6 & 0.2 \end{bmatrix} \quad \sigma(z) = \text{ReLU}(z)$$

$$z_1 = W_1 x = \begin{bmatrix} 0.43 \\ 0.25 \end{bmatrix} \quad h_1 = \sigma(z_1) = \begin{bmatrix} 0.43 \\ 0.25 \end{bmatrix}$$

$$z_2 = W_2 h_1 = \begin{bmatrix} 0.6 & 0.2 \end{bmatrix} \begin{bmatrix} 0.43 \\ 0.25 \end{bmatrix} = 0.308 \quad \hat{y} = z_2 = 0.308 \quad y = 0.8$$

$$J = \frac{1}{2} (\hat{y} - y)^2 = 0.1202$$

$$\frac{\partial J}{\partial z_2} = \hat{y} - y = -0.492 \quad \frac{\partial J}{\partial W_2} = \frac{\partial J}{\partial z_2} h_1^T = -0.492 \begin{bmatrix} 0.43 & 0.25 \end{bmatrix}$$

$$\frac{\partial J}{\partial z_1} = \frac{\partial J}{\partial h_1} \odot \sigma'(z_1) \quad \frac{\partial J}{\partial W_1} = \frac{\partial J}{\partial z_1} x^T$$

$$\frac{\partial J}{\partial W_1} = \frac{\partial J}{\partial z_2} \frac{\partial z_2}{\partial h_2} \frac{\partial h_2}{\partial z_1} \frac{\partial z_1}{\partial h_1} \frac{\partial h_1}{\partial z_1} \frac{\partial z_1}{\partial W_1}$$

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